

**The Impact of a Professional Development Program on Teachers' Mathematical
Knowledge for Teaching, Instruction, and Student Achievement**

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Abstract

In this paper, we describe findings from a three-year evaluation of a well-developed mathematics professional development program that is commercially available on a wide-scale. The professional development is designed to improve teachers' mathematical knowledge for teaching and to enable them to elicit more student thinking and reasoning during mathematics lessons. Specifically, it focused on helping teachers (a) learn more mathematics, (b) understand how children learn math, (c) use formative assessment to develop insight into what specific students know and do not know, and (d) develop effective classroom instructional strategies that enable student problem solving. Participants included 105 4th and 5th grade teachers teaching in 19 low-income schools within a single district. Teachers were randomly assigned within schools either to a "business as usual" control group or to receive the professional development. The training consisted of a week-long summer institute and 4-6 in-service days during the school

year. The training was run by full-time trained associates. We find some limited evidence of positive impacts on teachers' mathematical knowledge for teaching, but no effects on instructional practice or student outcomes.

Keywords

teacher professional development, mathematics, elementary school, experimental design, teacher content knowledge

Efforts to improve mathematics instruction frequently rely on teacher professional development as a means to this end. In recent years there has been a heavy investment in programs that cover two aspects of instructional practice in particular: improving teachers' mathematical knowledge for teaching, and increasing the amount of student mathematical thinking and reasoning during instruction. Interest in these areas has been driven by current federal and state guidelines for mathematics instruction (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010; National Mathematics Advisory Panel, 2008), by prior research that has found consistent links between teachers' mathematical knowledge, improved instructional quality, and student outcomes (Hill, Rowan, & Ball, 2005; Hill et al., 2008; Hill, Kapitula & Umland, 2011; Baumert et al., 2010; Kersting, Givvin, Thompson, Santagata & Stigler, 2012), and by the perceived professional development needs of U.S. classroom teachers (Hiebert et al., 2005).

Evaluations of mathematics professional development programs designed to meet these needs, however, have returned mixed results. This is particularly true of programs that have the potential to operate at scale and thus have the capacity to reach large numbers of teachers (e.g., for positive results see Carpenter, Fennema, Peterson, Chiang, & Loef, 1989; Jacobs et al., 2007; Perry & Lewis, 2011; Roschelle et al., 2010; for null results see Dominguez, Nicholls & Storandt, 2006; Garet et al. 2011; Santagata, Kersting, Givvin, & Stigler, 2010). These findings suggest that more research is needed to help understand under what circumstances professional development yields positive impacts on teacher knowledge, instructional practice and student outcomes.

To this end, this paper describes the findings from an evaluation in which 105 teachers were randomly assigned within schools to either a “business as usual” control group or to receive professional development intended to improve their mathematical knowledge for teaching and instructional quality. The program, Math Solutions (www.mathsolutions.com) has provided professional development to over 600 districts in 48 states, reaching over 100,000 teachers. Math Solutions was chosen for the evaluation because it meets the criteria articulated in Desimone’s (2009) description of effective professional development program features; because an earlier, observational study had suggested promising results (Cohen & Hill, 2001); and because the program is commercially available to districts across the United States.

Background

Since 2010, a majority of U.S. states have adopted the Common Core State Standards. Central to this reform movement is the idea that teachers should offer students opportunities to learn not only more rigorous content but also more sophisticated ways of thinking and reasoning about that content (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010). The Common Core mathematics standards extend a longer set of reforms aimed at increasing the amount of student thinking, reasoning, and problem-solving in mathematics classrooms (NCTM, 1989; 2000). In 1985, California reformers enacted a new state mathematics framework that called for much more intellectually ambitious student work, more student engagement in mathematical discourse, and more teacher attention to students’ mathematical ideas and reasoning. Professional standards, formulated by the National Council of Teachers of Mathematics (NCTM), followed in 1989, emphasizing many of the same ideas. The

1989 NCTM standards provided a model for many state standards enacted during the 1990s and beyond.

Classroom implementation studies of these earlier reform efforts suggested that teachers' practice was slow to change in substantial ways (Cohen, 1990; Putnam, Heaton, Prawat, & Remillard, 1992; Peterson, 1990; Spillane & Zeuli, 1999; Weiss et al., 2003; Wilson, 1990; for a review of earlier pieces, see Knapp, 1997). Among other barriers, observers noted the difficulty of enacting ambitious instructional practice and the fact that many teachers lacked sufficient understanding of the content to support these new practices (see, e.g., Cohen, 2011; Heaton, 2000; Kennedy, 2005; Lampert, 2001). Scholars and policymakers soon argued that improvements in content knowledge would be an essential ingredient in instructional change, and that these improvements would lead to improved student outcomes (e.g., NCTM, 1991). During this time, scholars also began to differentiate between basic or "common" content knowledge, or the grade-level mathematical knowledge teachers must teach students, and "specialized" content knowledge, or the mathematical knowledge and skills uniquely needed by teachers in the conduct of their work, arguing that both were important in developing a strong teaching workforce (e.g., Ball, 1990).

Funded initially by state and federal money, the professional development marketplace responded. In 2000, California allocated funds for Math Professional Development Institutes (MPDIs) in which mathematicians and mathematics educators worked together to present teachers with content-focused, extended learning opportunities. These institutes served over 23,000 K--12 teachers between 2000 and 2003 (Hill & Ball, 2004). Federal programs followed,

including both National Science Foundation-funded and U.S. Department of Education-funded Math--Science Partnerships, which, like the MPDIs, encouraged collaborations between mathematicians and mathematics educators for the purposes of helping teachers learn content and improve their teaching. Universities and university faculty---frequent suppliers of programs to local districts---and districts themselves similarly redirected their efforts in this vein; teacher surveys conducted in both 2000 and 2012 suggest that most teachers reported at least some professional development emphasis on mathematics content and new teaching strategies (Banilower et al., 2013; Weiss, Banilower, McMahon, & Smith, 2001). At the same time, federal research dollars went to initiating or growing new content- and pedagogy-focused programs for teachers, including the Vermont Mathematics Institute, developed by Kenneth Gross and now commercially available as Intel Math; Developing Mathematical Ideas, created by Deborah Shifter and now also commercially available; and the MathForum, an online community of mathematicians and mathematics teachers that also offers curriculum and professional development.

Some of these scalable programs were rigorously evaluated. In one of the earliest examples of a randomized trial of mathematics professional development, Carpenter et al. (1989) reported that a month-long workshop focused on student learning strategies improved treatment-group teachers' attention to students' thinking and strategies, and improved those students' outcomes on researcher-developed but not standardized measures of achievement. This intervention, Cognitively Guided Instruction, later formed the basis for a professional development program that prepared teachers to capitalize on elementary students' use of algebraic reasoning. Teachers learned about this reasoning in school-based collaborative groups,

and then generated and experimented with student tasks based on that reasoning. In end-of-year interviews, students of these teachers had a better understanding of the equals sign and used more diverse strategies in solving problems than students of non-participating teachers (Jacobs et al., 2007). Perry and Lewis (2011) also reported positive impacts on student achievement when lesson study was combined with a fractions resource kit for teachers that provided them an opportunity to deepen their own understanding of fraction content. And Roschelle and colleagues (2010) showed that professional development focused on novel curriculum materials improved students' performance on a test of conceptual knowledge.

However, there have also been numerous studies that have yielded null results---including some from cluster-randomized trials of well-designed and/or scaled-up professional development programs. Santagata et al. (2010) developed and studied a program that featured repeated cycles of teachers studying mathematical content, analyzing videotaped lessons related to that content, and enacting those lessons in their classrooms. They found no main effects on teachers' mathematical knowledge, instruction, or student outcomes. Argentin, Pennisi, Vidoni, Abbiati, and Caputo (2014) also found no impact on student test scores from a widely-used Italian program that encourages teachers to experiment with and analyze lessons that contain applications of mathematics to real life. Finally, Garet et al. (2011) found minimal effects on teachers' knowledge and instruction, and no effects on student achievement from two large scale evaluations of professional development providers.

Five different research reviews have attempted to synthesize the information gleaned from the numerous evaluations of teacher professional development, in both mathematics and

other content areas, over the years, and to identify the specific aspects of these programs that appear to be most effective in producing learning gains. These syntheses have generally indicated positive impacts of professional development on student learning. Meta-analyses suggest that the effect sizes range between 0.21 (Blank & de las Alas, 2010) and 0.54 (Yoon et al., 2007) on student outcomes. These reviews also find that discipline-specific programs produce greater learning gains (Kennedy, 1999; Scher & O'Reilly, 2009), that programs that engage teachers in determining strategies for achieving desired goals feature better outcomes (Kennedy, 2016) and that longer-duration programs produce larger impacts on student outcomes (Yoon et al., 2007). Finally, a review by Desimone (2009) argues that effective professional development contains several key features: it is focused on subject matter content; teachers take an active role in the work; it is coherent with teacher beliefs and state and local policies; and it features the collective participation of teachers from a school. Although Desimone's list of key features is based largely on non-experimental, teacher self-report data (Garet, Porter, Desimone, Birman & Yoon, 2001) it has been widely cited as laying out the key core design features for professional development.

The Math Solutions Professional Development Program

This study extends this research literature by evaluating one widely used and scalable professional development program, which meets the criteria established by Desimone (2009). The program, Math Solutions, was founded by Marilyn Burns in 1984. The program's logic model is shown in Figure 1. As the Figure shows, the model is built on a foundation of district leadership support (superintendents, principals, curriculum directors and instructional coaches);

Math Solutions staff only engage in professional development in districts where there is strong leadership support for the professional development. Math Solutions staff then work with the district to tailor the professional development to meet needs identified by the district and to align with the district's math curriculum. Unlike many commonly used professional development programs, which are closely aligned with a curricular intervention, Math Solutions is not tied to any specific curricular program but is designed to adapt to any curriculum in use by a district.

Similarly, while many other large-scale programs often use a train-the-trainer approach, Math Solutions hand-selects and extensively trains full-time associates as staff members. Associates are located throughout the United States, which allows the program to easily reach interested schools and districts.

Four goals are central to the professional development program: helping teachers (a) learn more mathematics, (b) understand how children learn math, (c) use formative assessment to develop insight into what specific students know and do not know, and (d) develop effective classroom instructional strategies that enable student problem solving. The program emphasizes classroom strategies that involve engaging, high cognitive demand tasks where students can develop their own solutions to mathematical problems (often in collaboration with other students) and deepen their understanding of mathematical concepts. Teachers actively engaged with the content of the professional development through summer workshops and in-year follow up days, solving problems, discussing instructional approaches, and at times interviewing students and teaching demonstration lessons. Finally, as described in more detail below, the

program offered over 40 contact hours per year over a three-year period, and involved participation of groups of teachers from the same grade, subject and school.

Thus, the Math Solutions program meets all the requirements identified by Desimone (2009) and others as necessary for strong professional development (Garet et al., 2001; Hill, 2007; Desimone et al., 2001; Desimone, 2009). The program was specifically intended to improve teachers' mathematical knowledge for teaching and to enable teachers to elicit more student thinking and reasoning in classroom instruction. The activities in which teachers engaged were all designed to have an active learning component. Math Solutions staff worked with the district to design that content of the PD to help ensure coherence with the districts priorities (although, as will be discussed later as the district changed leadership and priorities over the course of the study, thus weakening the level of coherence in study years two and three). The program was of a sufficient duration (over 40 contact hours spread out over the course of a year) and was offered over a 3 year period of time to allow sufficient time for change, and involved participation of groups of teachers from the same grade, subject and school.

Method

The study was designed to answer the following primary research questions:

RQ1: Was the Math Solutions professional development program implemented with fidelity?

RQ2: How does Math Solutions professional development impact teachers' mathematical knowledge for teaching and the mathematical quality of their instruction?

RQ3: How does Math Solutions professional development impact students' mathematical achievement?

Below, we describe the data and methods used to answer these questions.

Sample and Setting

This study was conducted in one midsize school district serving a racially and socioeconomically diverse population of over 30,000 students across 46 school locations. Approximately 60% of the district's elementary schools are classified as Title 1. In addition, the area has experienced relatively high rates of student and teacher turnover due to a military presence in the surrounding community. This district has consistently performed below state averages in elementary mathematics.

The district's mathematics coordinator helped recruit 18 elementary schools to participate in the study, with a priority on recruiting schools that had (a) principals who were supportive of ambitious instruction and (b) varied student demographics. Eighteen principals agreed to participate in the study, allowing us to then recruit the fourth and fifth grade teachers slated to teach math in each of these schools during the first study year. Recruitment occurred in the summer and fall of 2010 and resulted in 88 teachers enrolling in the study (Cohort 1). In summer 2011, an additional 17 teachers (Cohort 2) were recruited to replace some of the 27 teachers who departed after the first year of the study, for a total sample of 105 teachers.

Of the 105 teachers originally recruited into the study, 52 taught fourth grade and 53 taught fifth grade. The number of teachers participating at each school varied due to school size,

interest, and the extent of subject specialization (e.g. in some schools one teacher taught multiple sections of math, while another did not teach any math). Table 1 shows descriptive statistics for these 105 teachers. These teachers were 72% white and 9% male, and 55% held graduate degrees. They had, on average, almost nine years of teaching experience. During pre-service training, around half (57%) had taken more than two college-level math courses, 35% had taken more than two math content courses for teachers, and 19% had taken more than two math methods courses. The baseline mathematical knowledge for teaching (MKT) scores of the teachers in the study were slightly below the national average for elementary teachers of all grades.

As described below, students of these teachers were assessed at the beginning and end of each study year. Descriptive statistics for the students who participated during the first study year are shown in Table 2. At baseline, 21% of the students were white (63% were black and 16% were of other race/ethnicities), 14% were receiving special education services and approximately 1% had previously been retained in grade. Students in the study and in the district more generally scored below the state average on district reading and mathematics assessments.

Study Design

After recruitment was finalized, teachers were randomized within their schools into treatment and control groups. Randomization resulted in 51 teachers (42 in Cohort 1 and 9 in Cohort 2) assigned to receive the professional development program (treatment group). The remaining 54 teachers (46 from Cohort 1 and 8 from Cohort 2) were assigned to a “business-as-usual” control group. There were more teachers with more than one college-level math content

course in the treatment group than in the control group. Otherwise, there were no statistically significant differences between teachers in the two groups (see Table 1). At baseline, there were slightly more white students than students of other ethnicities in the treatment group than in the control group and slightly fewer students receiving special education services in treatment than in control but no other statistically significant differences in the characteristics of the students in these two sets of classrooms at baseline (see Table 2).

As an incentive for participation, teachers in the control group were offered the opportunity to participate in district-designed science professional development in lieu of the mathematics professional development. The district received a lump sum financial contribution to hire an instructional coach to work with teachers on their science curriculum. The science professional development these teachers received was unrelated to the math professional development the teachers in the treatment group received and was unrelated to math instruction more generally.

This design employed teacher- rather than school-level random assignment. While school-level random assignment has several desirable features, including protection against treatment contamination within schools, the convenience of delivering the intervention at the school level and the conditions for collective participation (Desimone, 2009), the choice to randomize within school was made in order to minimize costs and ensure that at least some teachers in each school would receive the mathematics program. In addition, we assessed the risk of contamination as relatively low. Due to the focus on math content and emphasis on students as mathematical thinkers, as well as the previously documented difficulty changing instruction,

even in very intensive professional development programs (e.g., Garet et al., 2010; Santagata et al., 2010), we thought it was unlikely that control-group teachers would change substantially due to contact with treatment-group teachers. As described in detail below, we also assess the empirical evidence for contamination and find no evidence that the control group benefited from the mathematics professional development.

The Treatment

Teachers began their Math Solutions professional development with a four-day summer institute in August 2010, with additional summer institutes held in each subsequent summer through August 2012. The institutes were held at a district facility and attendance was strongly encouraged. In addition to the summer institutes, Math Solutions staff administered between four (Year 3) and six (Years 1 and 2) one-day, in-person sessions during the course of the school year.¹ In all, the program delivered over 40 contact hours per year over a three-year period, and involved participation of groups of teachers from the same grades, subjects, and schools. A separate training session was held each summer for school leaders and math coaches; these sessions featured mathematical problem-solving activities and observations of (and at times, participation in) the professional development sessions Math Solutions conducted with teachers.

Most professional development sessions experienced by teachers in this study followed the same basic template. Teachers typically started by working through a mathematical task or problem, often using multiple strategies or a concrete representation of the mathematical

¹ Distance learning and coaching were proposed by the Math Solutions team but not taken up by the teachers who were participating in the study.

operation or idea. Then, one of several activities followed: a discussion of how students might solve the task in different ways; an examination of concrete examples of student work, for example, by watching a video of students solving the task, or interviewing actual students from nearby classrooms; a demonstration of a teacher teaching the task to her class; a discussion of how the task exemplifies best practices in classroom instruction; or a discussion anticipating issues that might arise as students encountered the task.

Throughout the professional development, teachers were asked to reflect on their own changes in practice and to consider best instructional practices more generally. In addition, they were given books, often published by Math Solutions, with lesson ideas or detailed explanations of the mathematics that teachers might teach. Many of the activities that the Math Solutions staff used in training came from the books the teachers received. Finally, teachers were given opportunities to connect Math Solutions activities and materials to district pacing guides and the state test.

An activity from an August 2011 session illustrates how this occurred in practice. Teachers sitting in small groups were given a container of rice, tape, and three 5- by 8-inch index cards. The teachers were asked to make three shapes with the index cards---a circular cylinder, a triangular prism, and a square prism---by taping the two five-inch ends together with no gap or overlap. Once they had made the shapes, a Math Solutions staff member asked teachers to conjecture about which shape had the largest volume, or, alternatively, whether all shapes would hold the same amount. Many teachers argued that all the shapes would hold the same amount of

rice because they were the same height and the same distance around, having been made from the same-size index cards.

Teachers were then asked to test their hypotheses by actually filling the cylinder and prisms with rice and comparing the amount of rice used for each shape. Upon doing so, many were surprised that the circular cylinder actually held more than the other two shapes. In some groups, some teachers tried to recall and execute the formula for the volume of each shape, and in other groups, teachers tried to find the nets of the shapes; neither group made progress toward a reason for their outcome. The whole-group conversation that followed focused briefly on the mathematical reason for the cylinder's larger volume (the height was the same in all three shapes, but the areas of the bases differed) and then covered topics such as when to use the task during the school year, how long it should take to enact, and other desirable features of the task.

This activity sequence was typical of those we observed during both the summer and within-year sessions. The activities combined teachers collaboratively solving mathematics problems and discussing best practices in classrooms, ranging from mathematical practices (e.g., conjectures), to new forms of student assessment (e.g., interviews), to new kinds of classroom activities and pedagogical techniques. In some sessions, fourth and fifth grade teachers were separated into different rooms to work on grade-level specific content, or to make connections between district instructional guidance and Math Solutions materials.

Attrition Analysis

Table 3 shows the pattern of teacher attrition over the three years of the study. Teachers left the study for a variety of reasons, all unrelated to the study itself: they left study schools, left

teaching, switched grades, or stopped teaching math. Of the 88 Cohort 1 teachers who were randomly assigned in fall of the first year of the study, 82 (93%) participated in data collection at the end of Year 1. At the beginning of Year 2, an additional 17 teachers were randomly assigned, bringing the total study sample to 105; of these, 78 (74%) remained at the end of Year 2. Fifty-seven of the original 105 teachers (54%) remained in the sample at the end of Year 3.

Attrition can lead to biased estimates of program impacts if there is differential loss from the treatment or control groups. For example, if the strongest teachers leave the treatment group but their counterparts remain in the control group, the impact of the program will be understated. Attrition rates were low during the first study year. Overall attrition was 7% and differential attrition was 4% (treatment group attrition = 5%; control group attrition = 9%), so the estimated impacts from Year 1 are unlikely to be biased (What Works Clearinghouse, 2014). However, attrition rates rose in subsequent years (26% in Year 2; 46% in Year 3). In Year 2, differential attrition was still somewhat low (9% greater attrition among control than treatment teachers). In Year 3, however, the differential attrition rate rose to 20%, with attrition again higher among the control group, introducing a strong possibility of bias.

To explore the possibility of bias further, Table 4 shows descriptive statistics for the teachers who remained in the treatment and control groups at the end of study Year 2 and study Year 3. At the end of Year 2, there was only one statistically significant difference in the characteristics of the treatment and control teachers who remained in the study at the end of the second study year (more male teachers remained in the control group than in the treatment group). However, although not statistically significant, the magnitude of the differences between

the two groups was relatively large in some instances, including those who had prior professional development (74% vs. 64%), the number of math courses in college (66% vs. 50% had taken more than two) and the number of math content courses in college (47% vs. 27% had taken more than two). Findings were similar in Year 3. Again there was a statistically significant difference in the percentage of male teachers in the treatment versus control groups and in Year 3 there was also a statistically significant difference in the percentage of teachers who had taken more than two math content courses in college, with more treatment teachers than control teachers having taken more than two. In addition, some of the differences are relatively large in magnitude. For example the difference in baseline MKT scores between treatment and control teachers was 0.29 standard deviations, with control teachers having higher baseline MKT scores than treatment teachers.²

The bottom panel of Table 4 also shows the baseline student achievement scores of the students from these classrooms in Years 2 and 3. There were no statistically significant differences in the baseline achievement scores of students in treatment versus control classrooms.

Overall student attrition was 14% in study year 1, 7% in study year two and 16% in study year three. There was almost no differential attrition between the treatment and control group students (differential attrition was 0.002, 0.018 and 0.011 in years 1, 2 and 3 respectively). According to the What Works Clearinghouse (2014) this level of attrition is very unlikely to lead

² Mathematics coursework for teachers is uncorrelated with MKT scores in a nationally representative elementary sample; see Hill (2010).

to biased estimates. Each year teachers had a new set of students in their classrooms so attrition of students across years was not relevant.

Taken together, all this suggests that we can be relatively confident that the Year 1 findings provide unbiased estimates of program impacts. Evidence regarding potential bias introduced by attrition in Year 2 and Year 3 is more mixed, particularly in Year 3, and findings for these years should be interpreted with caution.

Data Collection

Data collection consisted of observations of Math Solutions sessions by study staff, teacher surveys, video-recorded observations of teaching practice, and student assessments. Additionally, the district supplied demographic and state test data for students participating in the study. Teachers received monetary compensation for general study participation and individual data collection activities.

Observations of Math Solutions sessions

Two of the authors observed 19 person-days of Math Solutions professional development, including a majority of days from the summer institutes and some school-year activities. Observers took notes and, when possible, also briefly discussed the day's activities with Math Solutions staff. We generated a description of the professional development activities (above) from this data, and also confirmed fidelity to the providers' general model for teacher development.

Teacher surveys

Teachers completed surveys at the beginning of the study (hereafter referred to as Time 1 [T1]), and in the spring of each of the three years that followed (T2, T3, and T4). Treatment teachers completed the T1 survey during the Math Solutions summer institute (in mid-August) and control teachers were delivered the T1 survey at their school in early September. At T2, T3, and T4, teachers provided data by completing two separate questionnaires in April and June of each year. At T2-T4, teachers in both treatment and control conditions completed the surveys at the same time.

Video-recorded teaching sessions

During the first two years of the study (T2 and T3), lessons taught by teachers in the treatment group were video-recorded; in the last year of the study (T4), both the treatment and control groups were recorded. At each time point, participating teachers provided video from six lessons; lessons averaged roughly one hour in length. Lessons were sampled in three blocks of two back-to-back taping days. To avoid repetition of content, teachers were asked to show different teaching units across blocks, and blocks were scheduled at least two weeks apart. Video capture was completed via a multi-camera unit developed by thereNow, a company that provides remote observation systems for the purposes of classroom research and professional development. School researchers set up the camera then left the classroom in order to minimize interference in the teacher's lesson. Teachers chose the day and time to be video recorded, but were asked not to record on days on which they would be giving an extended assessment. Parents were required to provide active consent for inclusion in this portion of the study.

Students whose parents did not provide consent (through refusal or abstention) were moved to seats out of camera view during recording sessions. Completed videos were scored by the research team using the instrument described below.

Student assessments

Two measures were used to capture student performance: the state standardized assessment and a researcher-developed instrument. The state standardized assessment in mathematics was administered each year between April and June under conditions typical of large-scale state tests. The researcher-developed mathematics assessment was administered in the fall (October) and spring (April) of each year of the study. Passive study consent forms were sent to parents at the beginning of each school year. Students whose parents refused study participation (fewer than 1.5 percent of the total sample in each study year, evenly split between treatment and control) were not tested and the district did not share their state test data. Study staff administered the assessment to students while the teacher was in the classroom, making appropriate accommodations for students with Individualized Education Programs (IEPs). Students could take up to 75 minutes to complete the assessment. More information about both tests is below.

District data

Following each academic year, district staff provided classroom-level demographic data, including gender, race/ethnicity, and special education status. As per the district's interpretation of Family Educational Rights and Privacy Act (FERPA) guidelines, individual-level student lunch eligibility data could not be transferred. District staff provided other student-level data,

including retention and end-of-year test scores for state reading and mathematics assessments for third, fourth, and/or fifth grades. Scores were reported for the year of the study and the one or two years prior, dependent on the student's grade.

Measures

Student outcomes

The state standardized assessment was developed by Pearson Education, and was changed significantly between the first and second year of the study with the stated intention that the new test would include more cognitively challenging items and be administered in a computerized format.³ A review of currently released mathematics items for the late elementary grades suggests that most items ask students to solve basic computation problems or relatively straightforward applications problems (e.g., Ellie sprints the 100-meter dash in 15.1 seconds; Jaden sprints the 100-meter dash in 14.05 seconds. How much faster is Jaden than Ellie?). For this assessment, the state reported reliabilities that ranged from 0.85 to 0.91, depending on grade, subject area, and type of administration (paper-and-pencil or computerized) (State Department of Education Technical Report, 2011).

The goal of using the researcher-developed assessment was to capture students' thinking about mathematical concepts, not just their ability to compute, and thus to provide an assessment more aligned with the general goals of mathematics reforms. Although the test was not specifically designed to align with the tasks and activities featured in the Math Solutions

³ All students received the same set of items in the computer format—i.e., there was no computerized adaptive testing.

professional development, it did share Math Solutions' view of the types of mathematics students should be learning. Multiple choice and free response items were designed to gauge, for instance, students' ability to understand and use representations, students' understanding of the equals sign and basic algebraic relationships, and students' understanding of fractions and decimals as quantities and as points on the number line. One item on the fourth grade test, for example, asked students to match the full partial products computation for 14×16 to a correct visual representation of this problem. The project-administered assessment covered number, operations, algebra, and geometry. To improve the coverage of grade-specific content within a given classroom, we administered four different test forms at each grade level. Students were randomly assigned to one of four test forms at pretest, and care was taken to make sure they did not receive the same form in the posttest administration. The reliabilities of the project-administered test forms ranged from a low of 0.82 to a high of 0.89.

Teacher self-report measures

Teacher questionnaires asked teachers to report on their instructional practices, self-assessed mathematics proficiency, and experiences in the professional development. These questions were used to assess overall satisfaction with the professional development and to compare the experiences of the program and control groups. Questions about instructional practice and self-assessed math proficiency were designed to assess whether teacher reports of their own behavior matched those from the observational measures. Factor analyses conducted within each year of data suggested strong internal validity; exploratory analyses typically indicated the factors as theorized, and these factors remained consistent across years of the study.

With few exceptions, reliabilities remained constant across years as well. The scales and reliabilities that arose from these survey items are:

Richness of the mathematics (TQ)

Teacher self-report of the presence of meaning-oriented mathematics in their instruction (e.g., why a procedure works, or demonstrating a procedure with manipulatives). Reliabilities were 0.86 at T1; 0.77 at T2; 0.69 at T3; and 0.74 at T4. The downward trajectory in reliability may relate to the fact that these items were embedded in a larger set that changed between T1 and T2.

Student participation in mathematical meaning-making and reasoning (TQ)

Teacher self-report of cognitive demand of student activities, including whether students were asked to solve complex problems, solve problems using multiple methods, explain their answers, or make sense of mathematical ideas. Reliabilities were 0.88 at T1; 0.87 at T2; 0.89 at T3; and 0.90 at T4.

Self-assessments of mathematical knowledge and teaching (TQ)

At T1, T3, and T4, teachers were asked to report on whether they felt they held the knowledge needed to teach proficiently at their grade levels and whether they considered themselves master mathematics teachers. Reliabilities were 0.86 at T1; 0.74 at T3; and 0.85 at T4.

Teachers were also asked at T2, T3, and T4 to report on the quality of their mathematics professional development during the prior year, including whether they learned content and

pedagogical strategies that were relevant to their practice, and whether they changed their teaching as a result of participation in the professional development. As this set of items was given to both treatment and control teachers, we did not name Math Solutions professional development. Thus, these reports are on all mathematics professional development experiences in the year prior to the survey. Reliabilities were 0.92 at T2; 0.93 at T3; and 0.98 at T4.

MKT measures

Teacher questionnaires included a set of items designed to measure teachers' knowledge in the areas of number and operations, geometry, and algebra. Only the number/operations and geometry data are considered here, owing to poor reliability for the algebra items. All items were developed by the Learning Mathematics for Teaching project (Hill, Schilling, & Ball, 2004), and can be characterized as measuring teachers' "common" content knowledge, or the grade-level mathematical knowledge teachers must teach students, and "specialized" content knowledge, or the mathematical knowledge and skills uniquely needed by teachers in the conduct of their work. Looking for patterns in student errors or understanding whether a non-standard approach to multiplying multi-digit whole numbers would work more generally are examples of this teaching-specific subject matter knowledge.

Two test forms were used for each subject area, and teachers were assigned to alternating test forms in each year of the study. Number/operations items were presented on each questionnaire, and algebra and geometry items alternated at each time point in order to reduce burden on teachers. Each test form included 25-30 MKT problems and roughly 60 items (some problems had multiple, nested items). Administrations were proctored by study staff, and most

teachers took roughly 45 minutes to complete the form. Teachers appeared to take completing of the MKT survey seriously at all time points, skipping fewer than two items at any given time point. We used off-the-shelf MKT forms, with reliabilities in our sample ranging from 0.64 to 0.82 across different scales and administrations. In our sample, teachers' scores were anchored to a larger, nationally representative sample of K--5 teachers. Negative scores indicate teachers (or groups of teachers) who were below the national average.

Observation measures

The Mathematical Quality of Instruction (MQI) instrument provided information about the quality of teachers' video-recorded mathematics instruction. As designed, the MQI includes four major dimensions: the Richness of the Mathematics (Richness); Student Participation in Meaning-Making and Reasoning (SPMMR); Teacher Errors and Imprecision (Errors); and Working with Students and Mathematics (Working with Students). *Richness* and *SPMMR* are conceptually analogous to the teacher self-report scales described above. *Teacher Errors* reports on the presence of mathematical errors introduced by the teacher during instruction, and *Working with Students* captures teacher use and remediation of students' mathematical ideas during the course of instruction. *SPMMR* is most closely aligned with Math Solutions' stated goals. Each dimension consists of between two and five items that identify specific behaviors.

Following scoring at the item level, raters were also asked to complete an overall assessment of the lesson on a 1--5 scale, with scores of 1--2 reserved for lessons that were mathematically problematic and scores of 4--5 reserved for lessons with both strong richness and student participation. Two trained raters were randomly assigned to each lesson and were blind

not only to treatment/control condition but also to district, as data from this study were scored alongside data from a much larger study of instruction.

Analysis

Our study was designed to answer research questions regarding both fidelity of implementation and the impact of the Math Solutions program on teachers' mathematical knowledge for teaching, the mathematical quality of their instruction and the achievement of their students. Here we describe the analyses conducted to address each of the research questions.

RQ1: Fidelity of implementation

We use qualitative data from observations of professional development sessions and discussions with the professional development providers and district leadership as well as attendance data collected at each professional development session to assess the degree to which the program adhered to the core elements outlined in program's logic model (Figure 1).

RQ2: Impact on teacher knowledge and instructional practice

Since this was a randomized study, testing the mean difference between treatment and control classrooms on our outcomes of interest would provide unbiased estimates of the impact of the program on teacher knowledge and instructional practice. However, we also included teacher-level covariates in our models, to account for possible chance differences between treatment and comparison teachers, and as a way to decrease the standard error in the estimate of program effects. Covariates include baseline levels of teacher knowledge (MKT scores), teacher

gender, education (BA or above), total years of teaching experience, and whether or not the teacher completed teaching methods or postsecondary mathematics coursework. Because teachers were nested within schools, schools are included as fixed effects in all models. At the end of each study year, for each teacher-level outcome measure (i.e., MKT scores, MQI scores, and survey measures), we estimated the following OLS model:

$$Y_j = \beta_1 T_j + \sum_{x=2}^x \beta_x COV_{xj} + \sum_{z=1}^z \beta_z SCH_{zj} + \varepsilon_j.$$

where,

Y_j is an individual teacher's score on the outcome of interest in a given year;

T_j is an indicator variable equal to 1 if the teacher is in the treatment group and 0 otherwise;

COV_{xj} comprises teacher-level covariates including the baseline measure of the outcome of interest;

SCH_{zj} includes indicator variables for each school in our sample, equal to 1 if teacher j teaches in that school or 0 otherwise; and

β_1 indicates the average impact of the program on the outcome of interest for teachers in our sample.

Thus, the coefficient on T (β_1) is the impact on our outcome of interest in each year of the study.

RQ3: Impact on student math achievement

To estimate treatment and control differences among students, we used a two-level hierarchical linear model with students nested within classrooms. Again, to improve model fit and eliminate chance differences in the groups at baseline, we included the following covariates: baseline achievement scores, race/ethnicity, gender, special education status, whether or not special accommodations were used as part of the assessment, and the mean baseline achievement of students within the class. Because students and classrooms were nested within schools, schools were included as fixed effects in all models. We estimated the following model:

$$\text{Level 1: } Y_{ij} = \pi_{0j} + \left(\sum_{s>0} \pi_{sj} X_{sij} \right) + \varepsilon_{ij}$$

where:

Y_{ij} is the student achievement score for student i from classroom j ;

π_{sj} is the regression-adjusted mean value of the outcome measure for classroom j ;

X_{sij} is the value of the s^{th} student-level covariate for student i from classroom j , and

ε_{ij} is the residual error for student i from classroom j , which is assumed to be independently and identically distributed.

$$\text{Level 2: } \pi_{0j} = \theta_0 + \theta_1 T_j + \left(\sum_{m>1} \theta_m Z_{mj} \right) + \sum_{m+1}^Z \theta_m SCH_{mj} + \mu_j$$

where:

θ_0 is the grand mean of the regression-adjusted outcome measure for the average control classroom;

T_j equals 1 for treatment classrooms and 0 otherwise;

θ_1 is the estimated impact of the treatment;

Z_{mj} is the m^{th} teacher-level covariate for classroom j ;

SCH_{mj} includes indicator variables for each school in our sample, equal to 1 if teacher j teaches in that school or 0 otherwise; and

μ_j is the residual error for classroom j , which is assumed to be independently and identically distributed.

As with the teacher-level analysis, we estimated the impact of the program on students at the end of each study year.⁴ Analyses were also conducted separately by grade.

Exploratory analyses

In addition to these main analyses, we also conducted two exploratory analyses with the goal of understanding whether the treatment impacted subgroups of teachers and classrooms differently. Based on the notion of “accumulated advantage” (Merton, 1968), we asked whether teachers with strong mathematical knowledge at baseline were more likely to profit from the program, particularly in the area of instructional improvement and student outcomes. We also

⁴ Students may have been exposed to teachers in the program for varying lengths of time. At most, students in the sample would have received two years of instruction from treatment teachers.

asked whether teachers who taught students with better baseline test scores were better able to enact Math Solutions instructional strategies and improve student outcomes than teachers with weaker students. The latter may occur, for instance, if teachers of higher-performing students found those students to be more open to the cognitively demanding mathematical tasks promoted by the intervention, or if teachers of higher-performing students felt less pressure to remediate student weaknesses and responded by using more innovative teaching techniques.

Missing data and other analytic considerations

As is true in most studies, we experienced some degree of missing data. We did not impute any missing outcome data; if data were missing on any outcome of interest for a particular teacher or student, that teacher or student was not included in that analysis. We did however, impute missing covariate data. In the case of covariate missing values, imputation was conducted using the school-by-grade block to which the teacher or student belonged. If more than 5% of observations were missing data for a given covariate, the model included an indicator variable specifying whether a student or teacher's data were imputed for this covariate.⁵

All results are presented as “intent-to-treat” estimates---that is, the impact of the program on all teachers in the sample, regardless of their level of participation. In this way we obtain unbiased estimates of the impact of *offering* the professional development to teachers. The intent-to-treat estimate answers an important policy-relevant question, namely, the impact of

⁵ Previous research finds that in group random assignment studies, all methods for accounting for missing data, including imputing a constant value with a missing value dummy, multiple imputation and maximum likelihood imputation result in low levels of bias (Puma et al., 2009).

implementing a professional development program with a group of teachers with variable levels of interest and commitment to it. However, these estimates do not indicate what the impact of the program would have been if all teachers had participated fully in the Math Solutions training.

Finally, the Cohort 1 and Cohort 2 teachers in our sample were different on a number of dimensions, including their total years of exposure to the professional development and their baseline MKT scores (Cohort 2 teachers started with higher levels of MKT). Therefore, as a sensitivity test, we also ran all of our models removing the Cohort 2 teachers to see if this affected the findings.

Results

In this section we describe the degree to which the program was implemented with fidelity and then review the impacts of the professional development program on teacher self-report measures, MKT measures, instructional quality measures, and student achievement. We regard the findings from the first year of implementation as the most rigorous, since there was little attrition during this year and fidelity of implementation was high. Although somewhat less rigorous due to attrition and lower attendance, Year 2 and Year 3 findings yield insights into the potential longer-term impacts of the program.

Fidelity of Implementation

We begin by assessing the fidelity of implementation. We take a structural approach to assessing fidelity and evaluate whether the core components of the Math Solutions program shown in Figure 1 were delivered as planned.

As shown in Table 5, fidelity of implementation was strong during the first year of the study, but declined in the subsequent two years. During the first year of the study, Math Solutions staff spent considerable time engaging with the district, assessing needs and developing the program in a way that would meet the needs of the district. However, after the district's math specialist retired at the end of the first year of the study, it was difficult to reengage with district leadership in the same way. The district was in a state of flux, with new state and district standards, new assessments and a new superintendent and as a result, the district staff appeared pulled in many different directions.

Professional development sessions in all three years were led by Math Solutions trained team members and, as detailed in Table 5, observations conducted by study team members suggest that the professional development was consistent with the description of the program goals and pedagogical methods in all three years. Attendance at the training was high during the first year of the program, when 35 of the 42 teachers (83%) randomly assigned to the treatment group attended the summer training and 90% (38 of 42) of the teachers participated in all six of the in-service training sessions that were held. However, attendance at the professional development dropped in the subsequent two years. During the second year, on average, only 69% of the teachers attended trainings; in the third year, only 55% of the teachers remaining in the treatment group attended. The week-long summer institute was shortened to 3 days in Year 2 and only 4 in-service days were offered in the last year of the study.

In the second and third study years, principal participation and engagement with the professional development also declined. During the first year of the study, most principals

attended the special session of the professional development designed to help them understand the training their teachers were receiving. During the third year, only two attended. In interviews, some teachers reported that principals were pulling students out during math for remediation so they missed the richer, concept-based, instruction they were trying to implement.

Teacher Self-Report Measures

Results for the teacher-level analysis are shown in Table 6. The first panel of the table shows the impact of the Math Solutions program on teachers' self-reported behavior and perceptions. There were no positive, statistically significant impacts of the Math Solutions program on teachers' assessments of their own mathematical knowledge and teaching, their students' participation in mathematical reasoning, or their perceptions of the richness of their math instruction across the three years of the study. In fact, in Year 2, teachers in the treatment group rated the *Richness of the Mathematics* in their classroom (i.e., the presence of meaning-oriented mathematics in their classroom) less favorably than the control group (effect size equal to -0.64), suggesting perhaps that teachers in the treatment group were more critical of their own instruction after participating in the professional development. The findings regarding teachers' own assessments of their instruction did not change when Cohort 2 teachers were removed.

Teachers in the treatment group were, however, significantly more likely in all three years to report that they had learned a great deal from their mathematics professional development experiences. This finding was echoed in teacher exit interviews, during which treatment-group teachers reported high levels of satisfaction with the program and with what they had learned.

MKT Findings

The second panel of Table 6 shows the impact of Math Solutions on teachers' number/operations and geometry MKT scores. There was a positive and statistically significant impact of the program on teacher MKT in number/operations in Year 2, and a positive (effect size equal to 0.32) but not statistically significant impact on this same measure in Year 3. The effect sizes here correspond to a 7% increase in the percent correct or answering about two more items correctly out of 28 total. There were no statistically significant differences between the two groups on the MKT geometry scores, although the effect size in Year 3 was positive and equal to 0.22. As described above, the geometry scales were not administered in Year 2.

As shown in Table 7, when we removed Cohort 2 teachers from the analysis, the statistically significant effect on the number/operations measure disappeared due to a reduction in the sample size, but the effect size shrank only modestly. Findings for geometry also remained the same when Cohort 2 teachers were removed. This provides some limited evidence that the Math Solutions program may have a small impact on the mathematical knowledge of the teachers who participated in the program with respect to number and operations.

MQI Findings

The final panel of Table 6 shows the impact of the Math Solutions program on the MQI measures. The means and standard deviations are typical of most other datasets containing the MQI (e.g. Blazar, 2015). Although the effect sizes on all four measures go in the expected direction---richness of instruction, working with students, and student participation in mathematical reasoning are all positive, while errors are negative---the differences are not

statistically significant. In addition, when Cohort 2 teachers were removed from the analyses (Table 7), the resulting impacts were very close to zero for all four MQI measures, suggesting that any positive impacts were driven almost entirely by the Cohort 2 teachers, who started the study with higher overall MKT scores than the Cohort 1 teachers. Finally, the MQI scores of the Cohort 1 treatment group teachers who persisted across all three study years did not increase, and in many cases decreased over this span of time. Means on Student Participation in Meaning-Making and Reasoning (SPMMR) at T2, T3 and T4 for these 24 teachers were 1.31, 1.22 and 1.13 in years one, two and three respectively, indicating a decline in the average quality ratings given to these teachers by coders over time. Findings were similar for ratings of Richness of Mathematics (Richness) and Working with Students. Thus, there is little evidence that the Math Solutions program had an impact on teachers' mathematical quality of teaching.

Student Achievement Findings

Table 8 shows the impact of having a teacher who participated in the Math Solutions program on two different measures of students' mathematics achievement---the study-administered assessment and the end-of-year assessment given by the state. There was no impact on student mathematics achievement in any year for either measure. These findings did not change when we removed Cohort 2 teachers from our analyses.

Exploratory Analyses

Finally, as described above, we also conducted two exploratory analyses to assess whether the treatment impacted subgroups of teachers and classrooms differently. We explored whether teachers with stronger mathematical knowledge at baseline (defined as having baseline

numbers and operations MKT scores above the 50th percentile) showed greater increases in the mathematical quality of their instruction or greater improvements in student outcomes as a result of the professional development. We also explored whether teachers who taught students with better baseline test scores (i.e. with average incoming student achievement scores on the study administered assessment above the 50th percentile) were better able to enact Math Solutions instructional strategies and improve student outcomes than teachers with weaker students.

We found no relationship between teachers' incoming MKT scores and the impact of the professional development on their MQI scores or the achievement of their students on either the state assessment or the study administered assessment. Thus, teachers with higher MKT scores did not appear to benefit more from the professional development than teachers who had lower levels of MKT at the start of the intervention. We also found no interaction effects between the professional development and measures of classroom-level incoming student achievement on either MQI scores or student achievement. However, we had somewhat limited power to assess impacts for both of these subgroups.

Evidence of Contamination

Contamination, in which the treatment group shares information or materials with the control group, is one potential concern in studies in which teachers are randomly assigned within schools. The proximity of teachers in the same school and group lesson planning can facilitate the sharing of materials and ideas. Similarly, in this study, school leaders had exposure to the training and might have shared information they learned with teachers from their schools who were in the control group. If such contamination were to occur, impact estimates would be

understated because there would be changes in the knowledge and behavior of the control group as well as the treatment group. However, there is no indication that contamination occurred in this study.

On surveys teachers were asked about their professional development experiences as well as their collaboration with teachers in the PD. Among the control teachers, 94% indicated that they had never observed a Math Solutions instructor teach a lesson to students, 97% said they had never had a Math Solutions instructor observe their own instruction. There were six teachers in the control group who indicated that they had consistently planned lessons with a coach or teacher participating in the MS training or that they had received materials from a Math Solutions coach. However those six teachers do not look different than the other control teachers in the sample on a) their MQI scores or b) the achievement level of their students. Furthermore, they scored somewhat lower on the MKT than control teachers who did not report regularly collaborating with a MS teacher or coach, all of which suggests that the collaboration did not increase their mathematical knowledge for teaching or change their instructional practice.

At the same time, the MKT measures for the control group remained flat over the three years of the study, while the MKT scores for the teachers in the treatment group grew. For example, among treatment group teachers who remained in the study for all three years, scores on the MKT numbers and operations assessment grew by 0.25 standard deviations while control group teachers only gained 0.01 standard deviations. This suggests that mathematical knowledge for teaching was not being shared between the two groups. Finally, while we do not have longitudinal data on the MQI scores of the teachers in the control group, the MQI scores of the

treatment and control group teachers did not differ in the third year of the study. Furthermore the MQI scores of the treatment group teachers remained relatively flat and in many cases declined over the three year study period. Thus, the only way in which contamination could explain the null results on MQI scores is if the sharing of materials and the contamination from school leaders and principals was *more* effective in improving instructional practice for control group teachers than for the treatment teachers who actually participated in the Math Solutions PD sessions.

Discussion

The findings reported here are disappointing, given the time and energy devoted by Math Solutions staff and teachers to the professional development and the apparent value that teachers saw in their experience. Overall, we found some very limited evidence that the professional development may have had a small positive impact on teachers' mathematical knowledge for teaching, but this finding was positive and statistically significant only for one measure in one year. We found no changes in teachers' measured instructional practices or in the achievement of their students. These findings were observed despite the fact that the program met almost all of the key criteria for professional development effectiveness proposed by Desimone (2009).

We look to the theory of change outlined in Figure 1 for possible insight into these findings. As already noted above, the implementation of the program, particularly in the first year of the study was quite strong, and exhibited a high degree of fidelity to the program model. Math Solutions obtained strong buy-in from the district, worked with them to assess their needs and developed a professional development plan that was well-aligned to district needs and curriculum. The one-

week summer institute and the six in-service days were implemented as planned with high rates of participation, and the observations of the professional development sessions indicated that the topics covered were those outlined in the theory of change.

However, in study years two and three, district leadership support for the program waned considerably. At the end of the first year of the study the district's math coordinator, who was the primary champion of the program, retired. At the same time, the state and district mathematics standards and pacing guides were updated, the state assessment changed from paper-and-pencil to a computerized format, and a new superintendent was hired (the second of three during the course of the three-year study). As a result, it was more difficult to schedule in-service trainings, there was little support from district leadership to encourage teacher attendance at trainings and the priorities of the district were not as closely aligned with the professional development as it originally had been designed. Principal support for the professional development dropped substantially in these years as well, with only two principals attending the professional development sessions designed for them in the third year. Thus, one possible explanation for the findings is that in study years two and three, one of the foundational components of the Math Solutions model, district leadership support, was no longer in place.

This hypothesis is supported by others who have encountered similar phenomena. Santagata and colleagues reported a similar issue in their study (Givvin & Santagata, 2011; Santagata et al., 2010) and others have written extensively about the importance of situating professional development in settings that provide a coherent message and incorporating regular meetings with school and district leadership (see Sanders, Goldenberg, & Gallimore, 2009). For

instance, the professional development program known as eMINTS (Meyers, Molefe, Dhillon, & Zhu, 2015) built extensive work with principals -- even training them on a program-aligned observation rubric -- into their successful program. Both Wanless and colleagues (Wanless, Patton, Rimm-Kaufman & Deutsch, 2013) and Matsumura and colleagues (Matsumura, Sartoris, Bickel & Garnier, 2009) identified principal support as a key factor in their program's implementation.

Another possible explanation for the findings can also be found in Figure 1. While the content delivered as a part of the program (shown in Column 5 in the Figure) was covered as planned during the professional development, this content coverage did not lead to changes in instructional practice (column six in the Figure). It is possible that the nature of the professional development itself was not sufficient to change instructional practice. Research that has examined teacher participation in professional development suggests that professional development that explicitly links knowledge and practice is more likely to be effective (e.g. Lewis, 2009). Practices such as rehearsals, lesson analysis and guided reflection can help to solidify the link between knowledge and instructional practice (Kazemi, Franke, & Lampert, 2009). Coaching can also be an effective way to ensure these linkages are taking place. Among professional development programs that have been rigorously evaluated, those that have a coaching element have been relatively successful (e.g., Allen et al., 2011; Biancarosa & Bryk, 2011; Campbell & Malkus, 2013; Powell & Diamond, 2011; Ramey et al., in press). Kennedy (2016) also argues that programs that provide teachers broad goals and then facilitate opportunities for teachers to work together or with coaches to achieve those goals are more likely to be successful. These types of activities were not the primary focus of the Math Solutions

professional development and the professional development did not include a coaching component. Adding more of these types of activities might have led to more positive outcomes.

Programs in which curriculum material and professional development are combined also tend to be more successful (e.g., Assel, Landry, Swank, & Gunnewig, 2007), while stand-alone efforts to improve the quality of instruction are less so. For example, Gersten and colleagues' examination of teacher study groups found effects on instructional practice but not on student outcomes (Gersten et al., 2010). Santagata and colleagues' study (2010) of modules that enabled teachers to reflect on videotaped instruction and Garet's studies (Garet et al., 2008, 2010, 2011) of content-focused professional development in reading and mathematics also found null effects. Although the Math Solutions program did contain materials for teachers to use, the district did not generally support teachers' use of those materials, preferring to ask teachers to use their district-approved text, at least in the first year of the program.

Finally, the Math Solutions program may not have been a good fit for the needs of the teachers in this district. Prior research shows that among teachers who lack a strong foundation in mathematical knowledge for teaching, the implementation of new teaching methods can be problematic (Cohen, 1990; Hill et al., 2008; Hill & Charalambous, 2012). A number of teachers in our sample started the study with relatively low levels of mathematical knowledge for teaching. So, while the teachers may have been positive about their experiences with Math Solutions PD, the mathematics content and strategies that Math Solutions was designed to teach may have been beyond the current level of knowledge and skill of some of the teachers, making it difficult to implement the suggested strategies effectively.

Thus, the content of the Math Solutions professional development may have been sufficient to transform teachers' belief systems about instruction as demonstrated through teachers' survey responses, interview comments and to some extent their MKT scores, but insufficient in terms of providing the tools and supports for putting those beliefs into practice in their classroom. Substantial change may only come when teachers have the mindset, but also the materials and ongoing support to implement what they have learned.

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References

- Allen, J.P., Pianta, R.C., Gregory, A., Mikami, A.Y., & Lun, J. (2011). An interaction-based approach to enhancing secondary school instruction and student achievement. *Science*, 333(6045), 1034-1037.
- Argentin, G., Pennisi, A., Vidoni, D., Abbiati, G., & Caputo, A. (2014). Trying to raise (low) math achievement and to promote (rigorous) policy evaluation in Italy: Evidence from a large-scale randomized trial. *Evaluation Review*, 38(2), 99–132.
- Assel, M., Landry, S.H., Swank, P.R., & Gunnewig, S. (2007). An evaluation of curriculum, setting, and mentoring on the performance of children enrolled in pre-kindergarten. *Reading and Writing: An Interdisciplinary Journal*, 20, 463–494.
- Ball, D. L. (1990). The mathematical understandings that prospective teachers bring to teacher education. *The Elementary School Journal*, 449-466.
- Banilower, E. R., Smith, P. S., Weiss, I. R., Malzahn, K. A., Campbell, K. M., & Weiss, A. M. (2013). *Report of the 2012 National Survey of Science and Mathematics Education*. Chapel Hill, NC: Horizon Research.
- Baumert, J., Kunter, M., Blum, W., Brunner, M., Voss, T., Jordan, A., ... & Tsai, Y. M. (2010). Teachers' mathematical knowledge, cognitive activation in the classroom, and student progress. *American Educational Research Journal*, 47(1), 133–180.
- Biancarosa, G. & Bryk, A.S. (2011). Efficacy of literacy collaborative professional development: A summary of findings. *Journal of Reading Recovery*, 10(2), 25-32. Retrieved from:

http://readingrecovery.org/images/pdfs/Journals/JRR/Vol10_No2_Spring-2011/JRR_10.2_Contents.pdf

Blazar, D. (2015). Effective teaching in elementary mathematics: Identifying classroom practices that support student achievement. *Economics of Education Review*, 48, 16-29.

Campbell, P. F. and Malkus, N. N. (2013). The mathematical knowledge and beliefs of elementary mathematics specialist-coaches. *ZDM*, 46(2), 213-225.

Carpenter, T. P., Fennema, E., Peterson, P. L., Chiang, C. P., & Loef, M. (1989). Using knowledge of children's mathematical thinking in classroom teaching: An experimental study. *American Educational Research Journal*, 26(4), 499–531.

Cohen, D. K. (1990). A revolution in one classroom: The case of Mrs. Oublier. *Educational Evaluation and Policy Analysis*, 12(3), 311–329.

Cohen, D. K. (2011). *Teaching and its predicaments*. Cambridge, MA: Harvard University Press.

Cohen, D. K. & Hill, H. C. (2001). *Learning policy: When state education reform works*. New Haven, CT: Yale University Press.

Desimone, L. M. (2009). Improving impact studies of teachers' professional development: Toward better conceptualizations and measures. *Educational Researcher*, 38(3), 181-199.

Desimone, L. M., Porter, A. C., Garet, M. S., Yoon, K. S., & Birman, B. F. (2002). Effects of a professional development on teachers' instruction: Results from a three-year longitudinal

- study. *Educational Evaluation and Policy Analysis*, 24(2), 81-112. Retrieved from: <http://www.jstor.org/stable/3594138>.
- Dominguez, P. S., Nicholls, C., & Storandt, B. (2006). Experimental Methods and Results in a Study of PBS TeacherLine Math Courses. *Hezel Associates (NJ1)*.
- Powell, D. R. & Diamond, K.E. (2011). Improving the outcomes of coaching-based professional development interventions. In D. K. Dickinson and S. B. Neuman (Eds.), *Handbook of Early Literacy Research: Vol. 3* (pp. 295–307). New York, NY: Guilford.
- Garet, M. S., Cronen, S., Eaton, M., Kurki, A., Ludwig, M., Jones, W., Uekawa, K., Falk, A., Bloom, H., Doolittle, F., Zhu, P., and Szejnberg, L.. (2008). *The impact of two professional development interventions on early reading instruction and achievement*. (NCEE 2008-4030). Washington, DC: National Center for Education Evaluation and Regional Assistance, Institute of Education Sciences, U.S. Department of Education.
- Garet, M., Porter, A., Desimone, L. Birman, B., & Yoon, K. (2001). What makes professional development effective? Analysis of a national sample of teachers. *American Education Research Journal*, 38(4), 915-945.
- Garet, M. S., Wayne, A. J., Stancavage, F., Taylor, J., Eaton, M., Walters, K., & Doolittle, F. (2011). *Middle school mathematics professional development impact study: Findings after the second year of implementation* (NCEE 2011-4024). Washington, DC: National Center for Education Evaluation and Regional Assistance, Institute of Education Sciences, U.S. Department of Education.

- Garet, M. S., Wayne, A. J., Stancavage, F., Taylor, J., Walters, K., Song, M., & Doolittle, F. (2010). *Middle school mathematics professional development impact study: Findings after the first year of implementation* (NCEE 2010-4009). Washington, DC: National Center for Education Evaluation and Regional Assistance, Institute of Education Sciences, U.S. Department of Education.
- Gersten, R., Dimino, J., Jayanthi, M., Kim, J. S., & Santoro, L. E. (2010). Teacher Study Group: Impact of the Professional Development Model on Reading Instruction and Student Outcomes in First Grade Classrooms. *American Educational Research Journal*, 47(3), 694-739.
- Givvin, K. B., & Santagata, R. (2011). Toward a common language for discussing the features of effective professional development: The case of a US mathematics program. *Professional Development in Education*, 37(3), 439-451.
- Heaton, R. M. (2000). *Teaching mathematics to the new standard: Relearning the dance* (Vol. 15). New York, NY: Teachers College Press.
- Hiebert, J., Stigler, J. W., Jacobs, J. K., Givvin, K. B., Garnier, H., Smith, M., ... & Gallimore, R. (2005). Mathematics teaching in the United States today (and tomorrow): Results from the TIMSS 1999 video study. *Educational Evaluation and Policy Analysis*, 27(2), 111–132.
- Hill, H. C. (2007). Learning in the teaching workforce. *The Future of Children*, 111-127.

- Hill, H.C. (2010). The nature and predictors of elementary teachers' Mathematical Knowledge for Teaching. *Journal for Research in Mathematics Education*, 41 (5), 513-545.
- Hill, H.C. & Charalambous, C. Y. (2012). Teacher knowledge, curriculum use, and quality of instruction: Lessons learned and open issues. *Journal of Curriculum Studies*, 44(4) 559-576.
- Hill, H. C., Kapitula, L. R. & Umland, K. L (2011). A validity argument approach to evaluating value-added scores. *American Educational Research Journal* 48(3), 794-831.
- Hill, H. C., Blunk, M., Charalambous, C., Lewis, J., Phelps, G. C. Sleep, L. & Ball, D. L. (2008). Mathematical Knowledge for Teaching and the Mathematical Quality of Instruction: An Exploratory Study. *Cognition and Instruction*, 26, 430-511.
- Hill, H. C., Rowan, B., & Ball, D. L. (2005). Effects of teachers' mathematical knowledge for teaching on student achievement. *American Educational Research Journal*, 42, 371- 406.
- Hill, H. C., & Ball, D. L. (2004). Learning mathematics for teaching: Results from California's Mathematics Professional Development Institutes. *Journal of Research in Mathematics Education*, 35, 330–351.
- Hill, H. C., Schilling, S. G., & Ball, D. L. (2004). Developing measures of teachers' mathematics knowledge for teaching. *Elementary School Journal*, 105, 11–30
- Jacobs, V. R., Franke, M. L., Carpenter, T. P., Levi, L., & Battey, D. (2007). Professional development focused on children's algebraic reasoning in elementary school. *Journal for Research in Mathematics Education*, 38(3), 258–288.

- Kazemi, E., Franke, M. L., & Lampert, M. (2009). Developing pedagogies in teacher education to support novice teachers' ability to enact ambitious instruction. Paper presented at the Crossing divides: Proceedings of the 32nd annual conference of the Mathematics Education Research Group of Australasia (Vol. 1). Auckland, New Zealand.
- Kennedy, M. M. (2016). How does professional development improve teaching? *Review of Educational Research*.
- Kennedy, M. M. (2005). *Inside teaching: How classroom life undermines reform*. Cambridge, MA: Harvard University Press.
- Kennedy, M. M. (1999). Form and substance in mathematics and science professional development. *NISE Brief*, 3(2), 1–7. Madison, WI: National Institute for Science Education, University of Wisconsin.
- Kersting, N. B., Givvin, K. B., Thompson, B. J., Santagata, R., & Stigler, J. W. (2012). Measuring usable knowledge: Teachers' analyses of mathematics classroom videos predict teaching quality and student learning. *American Educational Research Journal*, 49, 568–589.
- Knapp, M. S. (1997). Between systemic reforms and the mathematics and science classroom: The dynamics of innovation, implementation, and professional learning. *Review of Educational Research*, 67(2), 227–266.
- Lampert, M. (2001) *Teaching problems and the problems in teaching*. New Haven, CT: Yale University Press.

Lewis, C. (2009) What is the nature of knowledge development in lesson study? *Educational Action Research*, 17:1, 95-110.

Matsumura, L. C., Sartoris, M., Bickel, D. D., & Garnier, H. E. (2009). Leadership for literacy coaching: The principal's role in launching a new coaching program. *Educational Administration Quarterly*, 45(5), 655-693.

Merton, R. K. (1968). The Matthew effect in science. *Science*, 159(3810), 56–63.

Meyers, C., Molefe, A., Dhillon, S. and Zhu, B. (2015). *The impact of eMINTS professional development on middle school teacher instruction and student achievement: Year 3 report*. Washington, D.C.: American Institutes for Research.

National Council of Teachers of Mathematics. Commission on Teaching Standards for School Mathematics. (1991). *Professional standards for teaching mathematics*. Washington, DC: Authors.

National Governors Association Center for Best Practices & Council of Chief State School Officers. (2010). *Common Core State Standards for Mathematics*. Washington, DC: Authors. Retrieved from http://www.corestandards.org/assets/CCSSI_Math%20Standards.pdf

National Mathematics Advisory Panel (2008). *Foundations for success: The final report of the National Mathematics Advisory Panel*, Washington, DC: U.S. Department of Education.

Perry, R. R., & Lewis, C. C. (2011). Improving the mathematical content base of lesson study summary of results. Retrieved September 25, 2015, from

<http://www.lessonresearch.net/IESAbstract10.pdf>

Peterson, P. L. (1990). Doing more in the same amount of time: Cathy Swift. *Educational Evaluation and Policy Analysis*, 12(3), 261–280.

Putnam, R. T., Heaton, R. M., Prawat, R. S., & Remillard, J. (1992). Teaching mathematics for understanding: Discussing case studies of four fifth-grade teachers. *Elementary School Journal*, 93, 213–228.

Puma, M. J., Olsen, R.B, Bell, S.H. & Price, C. (2009). *What to do when data are missing in group randomized controlled trials*. (NCEE 2009-0049). Washington, DC: National Center for Education Evaluation and Regional Assistance, Institute of Education Sciences, U.S. Department of Education.

Ramey, S.L, Ramey, C.T., Crowell, N.A., Grace, C., & Timraz, N. (in press). The dosage of professional development for early childhood professionals: How the amount, density, and duration of professional development may influence its effectiveness. In J.A. Sutterby, (Ed.), *Early childhood professional development: Research and practice through the early childhood educator professional development grant*. Boston, MA: The Emerald Group.

Roschelle, J., Shechtman, N., Tatar, D., Hegedus, S., Hopkins, B., Empson, S., ... & Gallagher, L. P. (2010). Integration of technology, curriculum, and professional development for

- advancing middle school mathematics three large-scale studies. *American Educational Research Journal*, 47(4), 833-878.
- Sanders, W. M., Goldenberg, C. N., & Gallimore, R. (2009). Increasing achievement by focusing grade-level teams on improving classroom learning: A prospective, quasi-experimental study of Title I schools. *American Educational Research Journal*, 46(4), 1006-1033.
- Santagata, R., Kersting, N., Givvin, K. B., & Stigler, J. W. (2010). Problem implementation as a lever for change: An experimental study of the effects of a professional development program on students' mathematics learning. *Journal of Research on Educational Effectiveness*, 4, 1-24.
- Spillane, J. P., & Zeuli, J. S. (1999). Reform and teaching: Exploring patterns of practice in the context of national and state mathematics reforms. *Educational Evaluation and Policy Analysis*, 21(1), 1-27
- State Department of Education Technical Report. (2011). *Available upon request from authors.*
- Wanless, S. B., Patton, C. L., Rimm-Kaufman, S. E., & Deutsch, N. L. (2013). Setting-level influences on implementation of the Responsive Classroom approach. *Prevention Science*, 14(1), 40-51.
- Weiss, I. R., Banilower, E. R., McMahon, K. C., & Smith, P. S. (2001). *Report of the 2000 National Survey of Science and Mathematics Education*. Chapel Hill, NC: Horizon Research.

Weiss, I. R., Pasley, J. D., Smith, P. S., Banilower, E. R., & Heck, D. J. (2003). *Looking inside the classroom*. Chapel Hill, NC: Horizon Research.

What Works Clearinghouse. (2014). *What Works Clearinghouse procedures and standards handbook* (Vol. 3.0). Retrieved from http://ies.ed.gov/ncee/wwc/pdf/reference_resources/wwc_procedures_v3_0_standards_handbook.pdf

Wilson, S. M. (1990). A conflict of interests: The case of Mark Black. *Educational Evaluation and Policy Analysis*, 12(3), 293-310.

Table 1 *Descriptive Statistics for Treatment and Control Group Teachers (Baseline)*

	All (N = 105)		Treatment (N = 51)		Control (N = 54)		% point difference
Variable	Mean	(SD)	Mean	(SD)	Mean	(SD)	(T--C)
Male	0.09	(0.29)	0.06	(0.24)	0.13	(0.34)	-0.07
White	0.72	(0.45)	0.73	(0.45)	0.70	(0.46)	0.04
Master's Degree or Higher	0.55	(0.50)	0.59	(0.50)	0.52	(0.51)	0.07
Fourth Grade	0.51	(0.50)	0.49	(0.50)	0.53	(0.50)	-0.04
Years of Teaching Experience	8.81	(8.13)	7.88	(8.67)	9.7	(8.06)	-1.81
Prior Professional Development	0.64	(0.48)	0.69	(0.47)	0.59	(0.50)	0.10
<u>College Coursework</u>							
More than Two Math Courses	0.57	(0.50)	0.61	(0.49)	0.53	(0.50)	0.08
More than Two Math Content Courses	0.35	(0.48)	0.45	(0.50)	0.25	(0.44)	0.19*
More than Two Math Methods Courses	0.19	(0.39)	0.20	(0.41)	0.18	(0.38)	0.03
							Effect Size
Baseline MKT scores	-0.08	(0.75)	-0.05	(0.80)	-0.11	(0.72)	0.06

Note. Exact sample sizes vary due to missing data; treatment/control group differences tested using simple linear regression with the descriptor (e.g., male) as the dependent variable and a treatment indicator as the independent variable; T--C = treatment/control group difference; MKT = mathematical knowledge for teaching.

* $p < .05$

Table 2 *Descriptive Statistics for Treatment and Control Students (Baseline)*

	All (N = 1523)		Treatment (N = 780)		Control (N = 743)		Difference
Variable	Mean	(SD)	Mean	(SD)	Mean	(SD)	(T--C)
Female	0.49	(0.47)	0.50	(0.47)	0.48	(0.47)	0.02
White	0.21	(0.38)	0.23	(0.40)	0.19	(0.37)	0.04*
Special Education	0.14	(0.13)	0.12	(0.12)	0.16	(0.33)	-0.04*
Previously Retained in Grade	0.01	(0.11)	0.01	(0.09)	0.02	(0.13)	0.01
Study Administered Math Scores	99.76	(9.57)	99.82	(9.65)	99.69	(9.49)	0.13
3 rd Grade District Reading Scores	453.06	(67.42)	455.24	(69.05)	450.78	(65.63)	4.46
3 rd Grade District Math Scores	469.63	(62.39)	470.76	(61.89)	468.44	(62.93)	2.32

Note. Exact sample sizes vary due to missing data; treatment/control group differences tested using simple linear regression with the descriptor (e.g., male) as the dependent variable and a treatment indicator as the independent variable; T--C = treatment/control group difference.

* $p < .05$

Table 3 *Pattern of Teacher Participation in the Study Over Time*

	T1 Baseline		T2 Year 1		T3 Year 2		T4 Year 3	
	T	C	T	C	T	C	T	C
Cohort 1	42	46	40	42	31	30	25	18
Cohort 2	9	8			9	8	8	6
Total	51	54	40	42	40	38	33	24
Study Participation Rate ^a	100/105 (95%)		82/88 (93%)		78/105 (74%)		57/105 (54%)	
Attrition Rate	5%		7%		26%		46%	
Differential Attrition Rate ^b			4%		9%		20%	

Note. T = treatment group; C = control group.

^aStudy participation rate indicates total number of teachers for whom we had any data available at that point in time (i.e., survey data, observation data, or student assessment data).

^bDifferential attrition rate was calculated by subtracting the attrition rate of the treatment group from the attrition rate of the control group and taking the absolute value.

Table 4 *Baseline Descriptive Statistics for Treatment and Control Teachers After Attrition*

	Year 2					Year 3				
	Treatment (N = 40)		Control (N = 38)		T--C Diff.	Treatment (N = 33)		Control (N = 24)		T-C Diff.
Variable	Mean	(SD)	Mean	(SD)		Mean	(SD)	Mean	(SD)	
Male	0.00	(0.00)	0.16	(0.37)	- 0.16* *	0.00	(0.00)	0.13	(0.35)	- 0.13 *
White	0.67	(0.47)	0.64	(0.49)	0.04	0.64	(0.49)	0.74	(0.45)	- 0.10
Master's Degree	0.59	(0.50)	0.52	(0.51)	0.07	0.54	(0.51)	0.64	(0.50)	- 0.10
Fourth Grade	0.45	(0.50)	0.57	(0.50)	-0.12	0.42	(0.50)	0.50	(0.51)	- 0.08
Years of Teaching Experience	8.47	(8.31)	10.47	(8.91)	-2.0	8.51	(8.47)	8.50	(6.82)	0.01
Prior Professional Development	0.72	(0.45)	0.57	(0.50)	0.12	0.70	(0.47)	0.61	(0.50)	0.09
Baseline MKT scores	-0.03	(0.86)	-0.04	(0.72)	0.01	-0.12	(0.79)	0.17	(0.73)	- 0.29
<u>College Coursework</u>										
More than Two Math	0.66	(0.48)	0.54	(0.50)	0.15	0.64	(0.49)	0.52	(0.51)	0.12

Courses										
More than Two Math Content Courses	0.47	(0.50)	0.27	(0.45)	0.20	0.45	(0.51)	0.13	(0.34)	0.31 *
More than Two Math Methods Courses	0.26	(0.45)	0.19	(0.40)	0.07	0.26	(0.44)	0.17	(0.39)	0.09
<u>Baseline Student Achievement</u>										
Study-Administered Math Scores	94.05	(10.72)	94.51	(11.01)	-0.46	93.36	(10.38)	93.97	(10.07)	- 0.61
3 rd Grade District Reading Scores	454.8 1	(73.36)	452.4 2	(71.64)	2.39	451.7 2	(67.16)	453.6 9	(58.40)	- 1.97
3 rd Grade District Math Scores	473.6 7	(75.79)	473.5 2	(69.13)	0.15	445.0 2	(71.53)	448.7 4	(62.83)	- 3.72

Note. Exact sample sizes vary due to missing data; T--C differences tested using simple linear regression with the descriptor (e.g., male) as the dependent variable and a treatment indicator as the independent variable; T--C Diff. = treatment/control group difference which in most cases indicated a difference in proportion. MKT = mathematical knowledge for teaching. MKT differences can be interpreted as effect size differences.

$*p < .05$,

$**p < .01$

Table 5 *Implementation Fidelity*

Fidelity Component	Year 1	Years 2 and 3
District Leadership Support	District support was explicitly given at the beginning of the study. The Superintendent came to one of the first PD sessions. Getting time off for teachers to engage in MS PD was not an issue. Most school principals and math coaches came to a special session of the PD to help them understand the training their teachers were receiving.	Change in District leadership. Support waned. Difficult to get days, locations, and substitute teachers for participants to engage in PD during the school year. More explicit pacing guides were published by the district that teachers were to follow and caps placed on length of math lessons in some schools. Few principals, but many math coaches, came to the special PD sessions set up for them.
Assess Needs and Develop PD to Meet Needs of the District	Met with District math specialist to determine the best mathematical topics and strategies to focus on during PD. The first PD session asked for feedback from teachers about particular topics they would like to better understand. Many of these topics were areas where students struggle on the state exam.	Continued to work with a new district math specialist and get feedback from teachers to determine the focus of the PD, but newly appointed math specialist was not as actively involved as previous math coordinator had been.
Training conducted by full-time, trained Math Solutions associate	Yes	Yes
1-week summer institute, 6-in-service trainings	Yes, 87% teacher attendance	Only three-day summer institute in Year 2 summer, only 4 teacher in-service days in Year 3. 69% teacher attendance in Year 2 and 55% teacher attendance in Year 3.

Math Content Knowledge Emphasized in PD	Teachers engaged in mathematical problem solving tasks multiple times a day during PD. Some tasks were difficult for many teachers. Tasks were followed by a discussion of the activity.	Teachers engaged in mathematical problem solving tasks multiple times a day during PD. Some tasks were difficult for many teachers. Tasks were followed by a discussion of the activity.
Insight into Individual Learners Emphasized in PD	Not explicitly emphasized as much as the other areas of focus, but instructional activities emphasized getting all students talking, and sharing their ideas.	Similar to year one, but teachers also shared insights they received as they tried to implement ideas from the training in their classrooms. These included different ways students think about material.
Understanding How Students Learn Math Emphasized in PD	Teachers received a MS published book with a section on “How Children Learn Math.” Focused discussion on this topic during PD. Topic was also emphasized in the PD activities. Many activities had teachers focus on aspects of particular tasks that would be easy or difficult for students and how to address student difficulties.	Slide from year one was used to re-emphasize how children learn mathematics. Principles shared the first year were used in activities. Many activities had teachers focus on aspects of particular task that would be easy/difficult for students and how to address student difficulties.
Effective Instructional Strategies Emphasized in PD	Teachers received multiple books published by MS with activities they could do with their students. Many of the activities were used with teachers during the PD.	Teachers received more activity books published by MS. Many of the activities were used with teachers during the PD. Teachers shared thoughts about particular lesson/strategies that they had tried in their classrooms.

Table 6 *Intent-to-Treat Impact Estimates of Math Solutions Professional Development on Knowledge and Instructional Practice: Full Sample*

	T2 (N = 80)			T3 (N = 72)			T4 (N = 56)		
Outcome Variable	β	SE	ES	β	SE	ES	β	SE	ES
<u>Teacher Self-Report</u>									
SPMMR-Survey	-0.13	0.14	- 0.18	-0.12	0.18	- 0.17	0.11	0.23	0.15
Richness-Survey	-0.21	0.16	- 0.28	-0.48*	0.20	- 0.64	-0.21	0.23	- 0.28
Self-Assessment of MKT	N/A			0.04	0.13	0.05	0.09	0.19	0.09
Learning from Professional Development ^a	0.74**	0.26	0.76	0.88**	0.25	0.91	1.95**	0.53	2.01
<u>Mathematical Knowledge for Teaching</u>									
Number and Operations	-0.04	0.17	- 0.04	0.40*	0.18	0.40	0.32	0.26	0.32
Geometry	0.12	0.18	0.12	N/A			0.22	0.29	0.22
<u>Observation of Instructional Practice</u>							(N = 57)		
Richness	N/A			N/A			0.02	0.02	0.10
Working with Students	N/A			N/A			0.04	0.03	0.20
Teacher Errors	N/A			N/A			-0.03	0.03	- 0.17
SPMMR	N/A			N/A			0.03	0.02	0.15

Note. Exact sample sizes vary slightly due to missing outcome data on some measures; covariates include corresponding baseline measures, education, gender, race, prior exposure to Math Solutions professional development, and college math course enrollment patterns; SE = standard error; ES = effect size (Cohen's D); MKT = mathematical knowledge for teaching; SPMMR = student participation in meaning making and reasoning.

^aOnly includes teachers who reported receiving some math professional development.

* $p < .05$,

** $p < .01$

Table 7 *Intent-to-Treat Impact Estimates of Math Solutions Professional Development on Knowledge and Instructional Practice: Cohort 1 Only*

	T2 (N = 80)			T3 (N = 56)			T4 (N = 40)		
Outcome Variable	β	SE	ES	β	SE	ES	β	SE	ES
<u>Teacher Self-Report</u>									
SPMMR-Survey	-0.13	0.14	-0.18	-0.06	0.23	-0.08	-0.14	0.28	-0.19
Richness-Survey	-0.21	0.16	-0.28	-0.53*	0.22	-0.71	-0.58	0.30	-0.77
Self-Assessment of MKT	N/A			-0.09	0.14	-0.11	0.00	0.29	0.00
Learning from Professional Development ^a	0.74**	0.26	0.76	0.85*	0.31	0.87	1.70*	0.70	1.75
<u>Mathematical Knowledge for Teaching</u>									
Number and Operations	-0.04	0.17	-0.04	0.37	0.20	0.37	0.36	0.39	0.36
Geometry	0.12	0.18	0.12	N/A			0.07	0.37	0.07
<u>Observation of Instructional Practice</u>							(N = 43)		
Richness	N/A			N/A			-0.00	0.03	0.00
Working with Students	N/A			N/A			0.02	0.04	0.10
Teacher Errors	N/A			N/A			0.00	0.03	0.00
SPMMR	N/A			N/A			0.02	0.02	0.10

Note. Exact sample sizes vary slightly due to missing outcome data on some measures; covariates include corresponding baseline measures, education, gender, race, prior exposure to Math Solutions professional development, and college math course enrollment patterns; SE = standard error; ES = effect size (Cohen's D); MKT = mathematical knowledge for teaching; SPMMR = student participation in meaning making and reasoning.

^aOnly includes teachers who reported receiving some math professional development.

* $p < .05$,

** $p < .01$

Table 8 *Intent-to-Treat Impact Estimates of Math Solutions Professional Development on Student Achievement*

		T2 (n = 1523 students) [N = 80 classrooms]			T3 (n = 1453 students) [N = 74 classrooms]			T4 (n = 963 students) [N = 56 classrooms]		
Outcome Variable		β	SE	ES	β	SE	ES	β	SE	ES
Study-Based Assessments	Grade 4	-0.01	0.60	0.00	1.30	0.95	0.10	0.41	1.16	0.03
	Grade 5	0.90	0.58	0.08	0.04	0.74	0.00	0.31	1.33	0.02
State Assessments	Grade 4	-2.14	4.85	-0.03	3.62	5.64	0.04	3.30	9.73	0.05
	Grade 5	-2.01	4.47	-0.02	1.61	5.98	0.08	-4.18	5.95	0.06

Note. Covariates include race, ESOL status, special education status, whether or not an accommodation was used, race, gender, prior achievement, and classroom-level achievement; SE = standard error; ES = effect size (Cohen's D).

* $p < .05$,

** $p < .01$

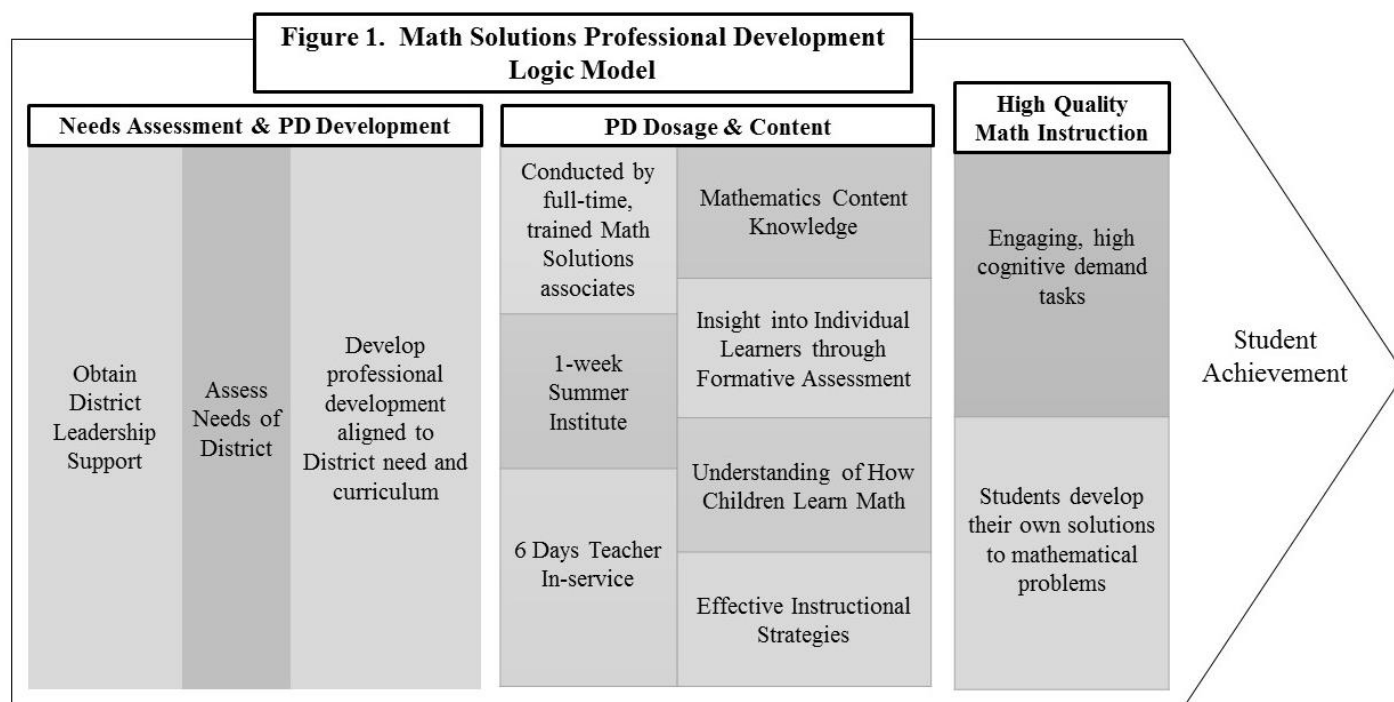


Figure 1 *Math Solutions Professional Development Logic Model*